Discrete Mathematics

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Contents

1 Logic **2**

1.1 Propositional Logic 2

1.2 Predicate Logic 4

2 Proofs **5**

2.1 Inference and Introduction 5

2.2 Methods and Strategy 7

3 Set Theory **9**

3.1 Definition and Relations 9

4 Functions **12**

4.1 Introduction 12

Chapter 1

Logic

1.1 Propositional Logic

Definition 1.1.1 (Proposition)**.** A **proposition** is a statement with presice meaning which have a truth value **true** or **flase**.

Definition 1.1.2 (Operators)**.** An **operator** or **connective** combines one or more operand express into a larger expresstion.

Table 1.1: Operator of propositions (Ordered by Precedence)

Formal Name	Nickname	Arity	Symbol	Precedence
Negation operator	NOT	Unary	$\neg$	1
Conjunction operator	AND	Binary	$\wedge$	2
Exclusive disjunction operator	XOR	Binary	$\oplus$	3
Disjunction operator	OR	Binary	$\vee$	4
Implication operator	IMPLIES	Binary	$\rightarrow$	5
Biconditional operator	IFF	Binary	$\leftrightarrow$	6

Definition 1.1.3 (Converse, Inverse, and Contrapositive)**.** The three statements of

an inplication are defined as followed:

$$\text{Converse: } q \to p$$

$$\text{Inverse: } \neg p \to \neg q$$

$$\text{Contropositive: } \neg q \to p.$$

The meaning of the contropositive of $\neg q \to \neg p$ is same as $p \to q$. Note that the "positive" in "contropositive" actually means "position" in English.

Example 1.1.4. $p \vee (\neg p \wedge q) \equiv (p \vee \neg p) \wedge (p \vee q) \equiv T \wedge (p \vee q) \equiv p \vee q$.

Definition 1.1.5 (Equivalence)**.** Compound proposition p is logically equivalent to compound proposition q, written $p \equiv q$, if the compound proposition $p \leftrightarrow q$ is a tautology.

Definition 1.1.6 (Tautology)**.** A tautology is a compound proposition that is always true.

For example, $p \vee \neg p$ is a tautology.

Definition 1.1.7 (Contradiction)**.** A contradiction is a compound proposition that is always false.

For example, $p \wedge \neg p$ is a contradiction.

Some equivence laws are listed as followed:

Theorem 1.1.8 (Some Logical Equivalences)**.**

1. Identity: $p \wedge T \equiv p \quad p \vee F \equiv F$.
2. Domination: $p \wedge F \equiv F \quad p \vee T \equiv T$.
3. Idempotent: $p \wedge p \equiv p \vee p \equiv p$.
4. Double negation: $\neg\neg p \equiv p$.
5. Associative: $(p \wedge q) \wedge r = p \wedge (q \wedge r)$. $\quad (p \vee q) \vee r \equiv p \vee (q \vee r)$.
6. Distribution: $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r) \quad p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$.

Theorem 1.1.9 (Logical Equivalences Involving Conditional Statements)**.**

1. $p \to q \equiv \neg p \lor q$.
2. $p \to q \equiv \neg q \to \neg p$.
3. $(p \to q) \land (p \to r) \equiv p \to (q \land r)$.
4. $(p \to r) \land (q \to r) \equiv (p \lor q) \to r$.
5. $(p \to q) \lor (p \to r) \equiv p \to (q \lor r)$.

Theorem 1.1.10 (Logical Equivalences Involving Biconditional Statements)**.**

1. $p \leftrightarrow q \equiv (p \to q) \land (q \to p)$.
2. $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$.
3. $p \leftrightarrow q \equiv (p \land q) \lor (\neg p \land \neg q)$.
4. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$.
5. $p \leftrightarrow q \equiv \neg(p \oplus q)$.

Theorem 1.1.11 (Logical Equivalences Involving Exclusive Disjunction)**.**

1. $p \oplus q \equiv (p \lor q) \land \neg(p \land q) \equiv (p \land \neg q) \lor (\neg p \land q) \equiv \neg(p \leftrightarrow q)$.

1.2 Predicate Logic

Definition 1.2.1 (Predicate logic)**.** A **predicate logic** is an extension of propositional logic that permits concisely reasoning about whole classes of entities.

For example, "For all $x \in \{1, 2, 3\}, x < 5$" is a proposition with x as the **object**, {1, 2, 3} as the **universess of discourse**, and $x < 5$ as the **predicate** or the **propositional function** mapping $x \to \{T, F\}$. Furthermore, "For all" is a kind of **quantifier,** which is defined as followed.

Chapter 2

Proofs

2.1 Inference and Introduction

Definition 2.1.1 (Proofs)**.** A **proof** is a correct(well-reasoned, logically valid) and complete(clear, detailed) argument that rigorously and undeniably establishes the truth of a mathematical statement.

Definition 2.1.2 (Theorems)**.** A **theorem** is a statement that has been proven to be true.

Definition 2.1.3 (Axioms, Postulates, Hypotheses, Premises)**.** An **axiom, postulate, hypotheses,** or **premises** is an assumption defining the structure about which we are reasoning.

Definition 2.1.4 (Rules of Inference)**.** A **rule of inference** is a pattern of logically valid deductions from hypotheses to conclusions.

Definition 2.1.5 (Lemmas)**.** A **lemma** is a minor theorem used as stepping-stone to proving a major theorem.

Definition 2.1.6 (Corollaries)**.** A **corollary** is a minor theorem proved as an easy consequence of major theorem.

Definition 2.1.7 (Conjectures)**.** A **conjecture** is a statement whose truth value has not been proven.

Definition 2.1.8 (Theories)**.** A **theory** is a set of all theorems that can be proven from a given set of axioms.

Now we talk about **inference rules** to help us construct proofs. Inference rules are tautologies allows us to proof the consequence true by antecedents, that is

$$\text{antecedent } 1 \wedge \text{antecedent } 2 \wedge \ldots \rightarrow \text{consequence},$$

or we write

$$\begin{array}{c} \text{antecedent } 1 \\ \text{antecedent } 2\ldots \\ \hline \therefore \text{ consequence} \end{array}$$

Here are several rules:

1. Rule of addition

$$\begin{array}{c} p \\ \hline \therefore p \vee q \end{array}$$

2. Rule of simplification

$$\begin{array}{c} p \wedge q \\ \hline \therefore p \end{array}$$

3. Rule of conjunction

$$\begin{array}{c} p \\ q \\ \hline \therefore p \wedge q \end{array}$$

4. Rule of modus ponens

$$\begin{array}{c} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

5. Rule of modus tollens (This kind of proof is called **indirect/ contrapositive proof**, example: a is rational, b is irrational, prove that $a + b$ is irrational.)

$$\begin{array}{l} \neg q \\ p \to q \\ \hline \therefore \neg p \end{array}$$

6. Rule of hypothetical syllogism

$$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$$

7. Rule of disjunctive syllogism

$$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

Another proof sumilar to indirect proof is **proof by contradiction**, that is, to show that p is true, we assume $\neg p$ is true, and then find a tautology that is $\neg p \to (q \wedge \neg q) \equiv F$, then we know that p must be true. (For example, the proof of "$\sqrt{2}$ is irrational")

2.2 Methods and Strategy

A proof of a statement of the form $\exists x P(x)$ is called an **existence proof**. If the proof demonstrate how actually find or construct a specific element a such that $P(a)$ is true, then it is called a **constructive proof**. Otherwise, it is called a **nonconstructive proof**.

Example 2.2.1 (An Constructive Proof)**.** Prove that for any integer $n > 0$, there exists a sequence of n consecutive composite integers.

Proof. Consider $(n+1)!+i+1$ for $i = 1, 2, 3, \ldots, n$. We know that $i+1|(n+1)!+i$. □

Example 2.2.2 (An Non-Constructive Proof)**.** Prove tha there are infinitely many prime numbers.

Proof. We show that $\forall n \, \exists p \, (p > n) \wedge (p \text{ is prime})$. Consider $n! + 1$, since $n! + 1 > 1$, so $n! + 1$ is either prime or composite. Using proof by case, if $n! + 1$ is prime, then we find $p = n! + 1 > n$. Otherwise, if $n! + 1$ is composite, it must have prime factor p. If $p \leq n$, then $n! + 1 \bmod p = 1$. A contradiction. Therefore, we have $p > n$. $\square$

Chapter 3

Set Theory

3.1 Definition and Relations

Definition 3.1.1 (Sets)**.** A **set** is a type of structure representing an unordered collection of zero or more distinct objects.

There are two kinds of notation representing sets. First, we can denot a set S by listing all of its elements in curly braces. For example $\{1, 2, 3\}$. Note that $\{1, 1, 2, 3\}$ and $\{3, 2, 1\}$ have the same meaning because elements of a set is unordered and distinct. Second, a set in the set builder notation represents a set of all elements satisfying a proposition over a domain. For example, $\{x|x > 0\}$ over $\mathbb{R}$.

Definition 3.1.2 (Equality of Sets)**.** Two sets are declared to be **equal** if and only if they contain excactly the same elements.

Some symbles represents spcial infite sets, including

1. Natual numbers: $\mathbb{N} = \{0, 1, 2, \ldots\}$.
2. Integers: $\mathbb{Z}$.
3. Real numbers: $\mathbb{R}$.
4. Empty set, or null: $\varnothing$.

Definition 3.1.3 (The Empty Set, or Null)**.** $\varnothing$ represents the **empty set**, or **null**, which is the unique set that contains no elements.

Note that $\varnothing = \{\} = \{x|F\}$. Now lets look about relations. A basic set relations is **member of**. We write $x \in S$ if x is an element or member of set S. Now we can define set equality in terms of $\in$ relation:

$$\forall S, T : (S = T) \leftrightarrow (\forall x : x \in S \leftrightarrow x \in T).$$

Also, we have

$$\neg\exists x : x \in \varnothing.$$

Definition 3.1.4 (Subsets, Supersets)**.** Let S, T be two sets. If $\forall x : x \in S \to x \in T$, we say that S is a **subset** of T, denoted $S \subseteq T$, and T is a **superset** of S, denoted $T \supseteq S$. Furthermore, $\nsubseteq$ and $\nsupseteq$ denotes the negative statements.

Theorem 3.1.5. For any set S, $S \subseteq S$.

Proof. $\forall x : x \in S \to x \in S$ is true. □

Theorem 3.1.6. For any set S, $\varnothing \subseteq S$.

Proof. Since $\forall x : x \in \varnothing$ is false, therefore, $\forall x : x \in \varnothing \to x \in S$ is true. □

Definition 3.1.7 (Strict/Proper Subets and Supersets)**.** Let S, T be two sets. If $S \subseteq T \wedge S \neq T$, we say that S is a **proper/strict subset** of T, denoted $S \subset T$, and T is a **proper/strict superset** of S, denoted $T \supset S$. Furthermore, $\not\subset$ and $\not\supset$ denotes the negative statements.

Definition 3.1.8 (Cardinality)**.** The **cardinality** of a set S, denoted $|S|$, is a measure of how many different elements S has.

For example, $|\varnothing| = 0, |\{1, 2\}| = 2$. If $|S| \in \mathbb{N}$, we say that S is **finite**, otherwise, we say that S is **infinite**.

Definition 3.1.9 (Power Sets)**.** Let S be a set, the **power set** of S

$$P(S) \coloneqq \{x|x \subset S\}.$$

Definition 3.1.10 (Cartesion Products)**.** For sets $A_1, A_2, \ldots, A_n$, their **Cartesian product**

$$A \times B \coloneqq \{(a_1, a_2, \ldots, a_n) | a_1 \in A_1 \wedge a_2 \in A_2, \ldots \wedge a_n \in A_n.\}$$

Definition 3.1.11 (Union)**.** Let A and B be sets. The **union** of the sets A and B is defined as

$$A \cup B \coloneqq \{x | x \in A \vee x \in B\}.$$

Definition 3.1.12 (Intersection)**.** Let A and B be sets. The **intersection** of the sets A and B is defined as

$$A \cap B \coloneqq \{x | x \in A \wedge x \in B\}.$$

Definition 3.1.13 (Disjoint)**.** Let A and B be sets. If

$$A \cap B = \varnothing,$$

we say that A and B are **disjoint.**

Definition 3.1.14 (Difference, Complement)**.** Let A and B be sets. The **difference** of A and B, or the **complement** of B with respect to A

$$A - B \coloneqq \{x | x \in A \wedge x \notin B\}.$$

The complement of the set A with respect to the universal set U, is called the **complement** of A, denoted by $\bar{A}$.

Chapter 4

Functions

4.1 Introduction

Definition 4.1.1 (Function)**.** Let A and B be nonempty sets. A **function** f from A to B is an assignment of exactly one element of B to each element o fA. We write

$$f(a) = b$$

if b is the unique elemtne of B assigned by the funciton f to the element a of A. If f is a function from A to B, we write $f : A \to B$.

Definition 4.1.2 (Domain, Codomain, Image, Preimage, Range, map)**.** If f is a function from A to B, we say that A is the **domain** of f and B is the **codomain** of f. If $f(a) = b$, we say that b is the **image** of a and a is a **preimage** of b. The **range**, or **image** of f is the set of all image of elements of A. Also, f **maps** A to B.

Definition 4.1.3 (Operator)**.** An n-ary **operator** over the set S is any function from the set of ordered n-tuples of elements of S, to S itself.

Definition 4.1.4 (One-To-One, Injections)**.** A function f from A to B is said to be **one-to-one**, or **injective**, or an **injection** if and only if

$$\forall a, b \in Af(a) = f(b) \to a = b.$$

Definition 4.1.5 (Onto, Surjections)**.** A function f from A to B is called **onto**, **surjective**, or a **surjection**, if and only if

$$\forall b \in B(\exists a \in A \quad f(a) = b).$$

Definition 4.1.6 (Bijections)**.** A function f is a **one-to-one correspondence,** or **bijective,** or a **bijection,** if it is both one-to-one and onto.

Definition 4.1.7 (Inverse Functions)**.** Let f be a one-to-one correspondence from the set A to the set B. The **inverse function** $f : B \to A$ of f is defined as

$$\forall b \in B\, f^{-1}(b) = a \to f(a) = b.$$

Definition 4.1.8 (Composition)**.** Let g be a function from the set $A$to the set B and let f be a function from the set B to the set C. The **composition** of the functions f and g denoted for all $a \in A$ by $(f \circ g)(a) = f(g(a))$.

Definition 4.1.9. The sets A and B have the same cardinality, that is $|A| = |B|$ if and only if there is a one-to-one correspondence from A to B.

Definition 4.1.10. If there is a one-to-one function from A to B, the cardinality of A is less than of the same as th ecardinality of B, that is $|A| \leq |B|$. If $|A| \leq |B|$ and A and B have different cardinality, then $|A| < |B|$.

Remark. It doesn't give separate meaning to $|A|$ and $|B|$ for infinite sets.

Definition 4.1.11 (Countable)**.** A set that is either finite or has the same cardinality of $\mathbb{Z}^+$ is called **countable.** A set that is not countable is called **uncountable.**

Example 4.1.12. The set of odd positive integers A is countable because we have the one-to-one correspondence $f : \mathbb{Z}^+ \to A, f(x) = 2x - 1$.

Theorem 4.1.13. If elements in a set S can be listed as a series, and for any elements, it also appears once in the seriec and can be found at a finite n-th term, then the set is countable.

Proof. Let the series presents as $s_1, s_2, \ldots$, then the function $f : Z^+ \to S, f(z) = s_z$ is a one-to-one correspondence. □

Theorem 4.1.14. If A and B are countable, then $A \cup B$ is also countable.

Proof. If A or B is a finite set, then it is either trivial or being the Hilbert's Grand Hotel. If A and B are both infinte sets, then let the elements in A are $a_1, b_2, \ldots$ and those in B are $b_1, b_2, \ldots$. We can construct a series $a_1, b_1, a_2, b_2, \ldots$ to list all elements in $A \cup B$. □

Theorem 4.1.15 (Cantor–Schröder–Bernstein Theorem)**.** If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.